

Hackathon 3

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September 18, 2026

Calculate the location of the Fermi level with respect to the conduction band for an n-type semiconductor having a doping of 10^{20} cm^{-3} in thermal equilibrium. Use an open-ended method for the calculation. Consider the convergence criterion to be the case when the update to the solution is less than $1 \mu \text{ eV}$ For Fermi-Dirac Statistics

$$\begin{aligned}n &= N_c F_{1/2}(\eta_c) \\p &= N_v F_{1/2}(\eta_v) \\ \eta_c &= -\frac{E_C - E_F}{k_B T} & \eta_v &= -\frac{E_F - E_V}{k_B T} \\ F_{1/2}(\eta) &= \frac{2}{\sqrt{\pi}} \int_0^\infty \frac{\sqrt{x}}{1 + \exp(x - \eta)} dx\end{aligned}$$

For Maxwell-Boltzmann Statistics

$$n = N_C \exp\left(-\frac{E_C - E_F}{k_B T}\right) \quad p = N_V \exp\left(-\frac{E_F - E_V}{k_B T}\right)$$

Print the update and residual at convergence as shown in the startup code.